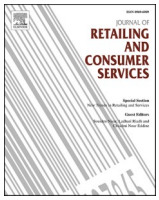
Journal of Retailing and Consumer Services 85 (2025) 104243



Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser





Improving consumer satisfaction through shopping app features: A Kano-based approach

Kathrin Sinemus [a,*], Stephan Zielke [a], Thomas Dobbelstein [b,c]

[a] University of Wuppertal, Gaußstr. 20, 42119, Wuppertal, Germany
[b] Durban University of Technology, 41/43 ML Sultan Road, 4001, Durban, South Africa
[c] Baden-Württemberg Cooperative State University, Marienplatz 2, 88212, Ravensburg, Germany

ARTICLE INFO

Handling Editor: H. Timmermans

Keywords:
Shopping app features
Consumer satisfaction
Kano model
Segmentation analysis

ABSTRACT

Most retailers offer a shopping app to their customers. Such shopping apps can include various features, such as product finding tools, wish lists, check and reserve, visual search, or augmented reality. For designing an app, retailers must know which features customers take for granted to prevent dissatisfaction and which features they appreciate to increase satisfaction. This study therefore applies the Kano model of customer satisfaction to analyze how a representative sample of 798 consumers categorize 25 app features as either must-be, attractive, one-dimensional, or indifferent. It further segments customers based on their tendencies to characterize the features and shows differences between these segments in several profiling factors, such as browsing orientation, app enjoyment, innovativeness, app usage frequency, and product involvement. This study extends existing literature by understanding how each of these features contributes to preventing dissatisfaction and increasing satisfaction and how this contribution differs between customer segments. Our study has in particular implications for designing shopping apps and targeting app design and communication of feature benefits to the identified segments.

1. Introduction

The world of retailing becomes more digital (Grewal et al., 2021). Consequently, consumers increasingly use different shopping channels in the customer journey and expect an omni-channel experience (Verhoef et al., 2015). Shopping apps can be a chance for retailers in this challenging context as they are a digital sales channel (Kim et al., 2017) that combines the online with the offline channel through app features such as a QR code scanner or click and collect (see Sinemus et al., 2022). Moreover, shopping apps offer retailers the opportunity to communicate personally with the consumer (Andrews et al., 2016; Natarajan et al., 2017; Park and Lee, 2017; Shankar et al., 2010) and to get in touch with them to strengthen the customer relationship (Peng et al., 2014; Taylor and Levin, 2014). Consequently, shopping apps represent an important channel for retailers.

Several retailers have a shopping app, however, the apps differ in the features they include. Some apps contain only core features such as a store finder, while others contain supplementary features that create an omni-channel experience (Inman and Nikolova, 2017), such as augmented reality functions or QR code scanners. It is important for retailers to know which features are important to consumers and which are less relevant, as due to limited budgets, retailers should focus on the features that customers value the most (Baier and Rese, 2020). Not every investment in the design of technologies or features guarantees the expected rate of return (Demko-Rihter and Ter Halle, 2015). For retailers, it will be therefore important to know, which features customers take for granted, which features are not necessarily expected but have the potential to delight customers and which features are not relevant for them. Furthermore, customers can differ in these expectations toward app features. Therefore, retailers must consider different customer segments regarding such expectations.

However, the literature on the aforementioned issues is scarce. Firstly, most studies analyze outcomes of shopping apps in general, but not on particular features. Studies focusing on features are often only conceptual and descriptive (Roggeveen and Sethuraman, 2020; Zhao and Balagué, 2015) or focus either on single app features or general feature groups (Sinemus et al., 2022). Secondly, these studies often do not differentiate whether these features are more likely to prevent

This article is part of a special issue entitled: Technology in Retailing published in Journal of Retailing and Consumer Services.

* Corresponding author.
E-mail address: Sinemus@wiwi.uni-wuppertal.de (K. Sinemus).

https://doi.org/10.1016/j.jretconser.2025.104243
Received 23 September 2024; Received in revised form 11 December 2024; Accepted 20 January 2025
Available online 21 March 2025
0969-6989/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

dissatisfaction as they are taken for granted or to increase satisfaction and delight customers. Two studies (Baier and Rese, 2020; Zhao and Dholakia, 2009) applied the Kano et al. (1984) approach for such purpose, but do not focus explicitly on shopping app features. Shopping apps particularly differ from websites through their easy access on the mobile phone (see Bellman et al., 2011), more options for personalization (Natarajan et al., 2017), and a wider range of integrated functions, as not all functions are integrated on the website, such as QR code scanners. Thirdly, previous research does not consider that customer segments differ in their tendency to take particular app features for granted or appreciate the presence of particular features. Baier and Rese (2020) applied such a segmentation, but focused on a broader range of technologies and not explicitly on app features and do not characterize segments by psychographic profiling factors.

We therefore address these research gaps by analyzing how consumers categorize 25 app features according to their potential to prevent dissatisfaction and to increase satisfaction. For this purpose, we apply the Kano approach that categorizes features as attractive, must-be, one-dimensional, or indifferent. Offering attractive features increases satisfaction, but not offering them does not result in dissatisfaction. Offering must-be features prevents dissatisfaction, but does not positively increase satisfaction. Offering one-dimensional features prevents dissatisfaction and increases satisfaction as well and indifferent features are unrelated to satisfaction. This categorization helps retailers to prioritize app features in the design of shopping apps. Furthermore, we use cluster analysis to identify customer segments and characterize the segments based on psychographic variables. The customer segmentation provides retailers with more insight into which features are important for different target groups.

This paper contributes to the literature by (1) categorizing a large range of 25 app features according to their potential to prevent dissatisfaction and increase satisfaction. We further show that this categorization depends on the penetration of features in retailers' shopping apps (frequently offered core features are more often taken for granted while selectively offered supplementary features potentially satisfy and delight customers). (2) We identify different consumer segments with a relatively stronger tendency to categorize features as must-be, attractive, or indifferent. We further show that these segments differ in their browsing orientation, app enjoyment, innovativeness, shopping app usage, and product involvement. Retailers can use results to design their shopping apps and advertise app features to the different customer segments. Hence, the novelty of our research lies in a categorization of a various core and supplementary app features according to their impact on satisfaction or dissatisfaction (in contrast to previous studies analyzing apps in general, single features, or website attributes) and a segmentation of consumers based on their feature categorization.

The next section introduces the 25 features, presents a literature review and the Kano approach as theoretical framework. Afterwards, we present propositions related to the impact of feature penetration and consumer psychographics on the categorization of features. We then analyze data from 798 consumers, segment them according to their categorization pattern and characterize the segments by psychographic profiling factors. The paper closes with a discussion, implications for theory and practice, and limitations.

2. Theoretical background

2.1. Shopping app features

Shopping apps "allow users to make 'shopping' transactions through their smartphones" (Kim et al., 2017, p. 58) and represent an additional sales channel alongside brick-and-mortar retail and traditional e-commerce (Peng et al., 2014; Taylor and Levin, 2014). Shopping apps offer benefits for both consumers and retailers. They can provide consumers with personalized content or recommendations (Tong et al., 2020), which can have a positive impact on consumer satisfaction (see e.g., Ampadu et al., 2022; Liang et al., 2007) or buying behavior (see e.g. Ampadu et al., 2022; Liu et al., 2021). In addition, users of shopping apps can create individual shopping profiles (Hoehle and Venkatesh, 2015). As a result, the purchasing process is very convenient for consumers and retailers receive data on consumers' purchasing behavior and preferences (Baier and Rese, 2020; Inman and Nikolova, 2017). Another advantage for retailers is that apps create greater exposure to consumers who see the app icon, which is often the company's logo (Bellman et al., 2011), multiple times on their smartphone. Through the app icon, consumers can more easily access personalized content and features without searching for and logging in to a website. Therefore, features might be more valuable for consumers, when they are integrated in a shopping app. Convenient access might be in particular important for features used in physical stores, such as a digital loyalty card or payment via app, as these features can only be used when consumers are logged in. Some features are further often available in shopping apps, but not or less on websites. These are, for example, augmented reality or a QR code scanner. Such features can lead to higher interactivity and novelty, which has a positive impact on customer experience and purchase intent compared to retailer websites (see e.g., Zare Ebrahimabad et al., 2024). In addition, Sinemus et al. (2022) have shown that app users are more satisfied with and loyal to retailers than non-users. These positive aspects of shopping apps show how important a well-designed app is.

Shopping apps are not only a sales channel, they also offer various features that support the consumer while shopping (Kim et al., 2016). Table 1 provides an overview of such features that retailers can integrate in their shopping apps. The literature has characterized and grouped these features based on different criteria, depending on the context of the studies and the particular focus on shopping apps or a more general focus on shopping processes. Sinemus et al. (2022), for example, assign shopping app features to three groups: features that support customers in the pre-purchase stage (product finding tools, augmented reality), their purchase transactions (loyalty card, shipment tracking), or their cross-channel behavior (availability checks, click and collect). Hence, their grouping considers stages of the purchase process and involved channels. With a more general focus, Ortlinghaus et al. (2019) discuss different cross-channel features, such as check and reserve, click and collect, and product availability checks, which combine online and offline channels. Other studies consider online and offline features separately. For example, Baier and Rese (2020) consider online and physical store technologies in the context of a multichannel retailer. They assign features such as a QR code scanner or a bonus card to the physical store and features such as a chat service or an online account to the online store. However, when assigning features, they point out that certain features have a cross-channel function, as already mentioned in Ortlinghaus et al. (2019), and thus connect channels.

As our study aims to analyze how particular features contribute to consumer satisfaction, we choose a distinction in core and supplementary features. We define core features as features that are frequently offered by retailers and thus seem to be important for them. Such features might be expected by customers, but offer little potential for differentiation. Supplementary features are, in contrast, features that are offered less frequently and only by single retailers. These features seem to be less relevant for the majority of retailers, but may offer potential for differentiation and satisfying customers. This raises the question which features a retailer should offer to create consumer satisfaction and to avoid consumer dissatisfaction.

2.2. Literature review

Previous research on omnichannel shopping has identified segments that use the mobile channel more frequently than other consumers. However, the studies do not consider that customer segments differ in their expectations regarding the features of shopping apps. For example, Herhausen et al. (2019) consider the usage of a mobile device in the

Table 1
Overview of app features.

Feature	Definition
Augmented reality	Using a smartphone camera to virtually try on glasses or items of clothing, for example, or have a television virtually projected into your living room
Chatbot	Using a chat window to communicate with a technical person from the retailer in their app and ask questions. The technical person can provide information on common issues, such as shipping.
Check and reserve	Reserving a product that is available in the stationary store online and pick it up later in the store (payment is made in the store)
Click and collect	Purchasing a product that is available in the stationary store online and pick it up later in the store (payment is made online in advance)
Click and meet	Making an appointment with the app to get advice at the local store
Comparison list	Comparison of two or more products in terms of product features, such as size, weight or image resolution, for example by comparing the product features in a table
Data preservation	Saving personal data such as name, address and payment method in the app
Electronic receipt	Viewing the receipts for both online purchases and in-store purchases. On-site purchases are recorded in the app via a digital customer card, for example
Get the look	Following influencers directly via the app and shop their outfits
Inbox	The inbox is comparable to an e-mail inbox. Receiving news directly in the app (similar to a newsletter)
Live chat	Communicate with a real employee of the retailer through the app, ask questions and get advice. This type of chat should not be confused with a chatbot (technical person)
Loyalty program	Participating in a loyalty program and receive benefits such as exclusive discounts or collecting points to redeem for free products
Online magazine	Contains editorial content, such as the latest trends, the latest technology, test reports or other areas in which the company is involved. The online magazine should not be confused with a catalog
Payment via app	Opening the app at the checkout in the stationary store, showing the customer card, for example, and pay using the payment method the customers have stored. The technical design can vary. But all designs have in common that customers can pay with the help of the shopping app
Personal product recommendations	Based on previous searches or purchases, personalized products are suggested to the customer that might suit/appeal them
Product availability check	Checking online whether a desired product is available in the stationary store
Product finding tools	Refining the search with product search aids. The product finding aids include, for example, filter functions or the search bar in which customers can enter a term
QR Code Scanner	Using the QR code scanner in the app and the smartphone camera to scan a QR or EAN code and the desired product will be displayed in the app
Re-availability e-mail	If a desired product is no longer available online, customers can get an e-mail sent to them as soon as the product is available again
Sharing product links	Send product links from the app directly to friends and acquaintances via, e.g. WhatsApp or email
Shipment tracking	Ordering a product in the app and viewing the shipment within the app (usually under "My orders"). This function works in the same way as shipment notifications from parcel service providers, but is visible in the app
Size finder	The app will recommend the right clothing size based on age, height and/or weight
Store Locator	Using the store locator to display the nearest stationary store
Visual search	Using a smartphone camera to take photos of products and display similar products in the app
Wish list	Saving products to a wish list

customer journey. The usage could refer to an app or the online browser on the smartphone. The authors do not find a pure mobile segment, however, they find that some segments use mobile devices for shopping more often than other segments. Schneider and Zielke (2020) include mobile devices in their segmentation of showroomers who visit physical stores for information search and then purchase online. They identify a mobile showroomer segment that uses mobile devices for searching information in-store and purchasing the product online during or shortly after the store visit. A study by Frasquet et al. (2019) aims to understand how customers use channels to complain about multichannel retailers. The authors identified a segment that uses mobile apps for complaints more often than others. In summary, previous literature has identified segments using mobile devices more frequently in their shopping process. However, the segmentation studies do not particularly focus on shopping apps and they do not analyze reasons for their usage. One reason could be the features offered by shopping apps, which can have a positive impact on consumer satisfaction (Sinemus et al., 2022) and thereby can lead to more frequent usage (see, for example, Hsiao et al., 2016; Iyer et al., 2018).

Regarding research on shopping apps, previous literature examines the outcomes of using mobile apps, such as satisfaction (Baek, 2013; Chang, 2015; Iyer et al., 2018; Maduku and Thusi, 2023; Sarkar and Khare, 2018; Sinemus et al., 2022), intention to (re)purchase (Baek, 2013; Iyer et al., 2018; Qin et al., 2021; Sarkar and Khare, 2018), or positive word of mouth (Sarkar and Khare, 2018). However, as stated by Li et al. (2020), the research on shopping app features has just emerged and only a few studies focus on shopping app features or the design of shopping apps (see Table 2).

Some studies provide an overview of (app) features, but are mostly descriptive. Roggeveen and Sethuraman (2020) suggest a conceptual framework for understanding retail technologies and identify ten areas that offer potential for future research on retail technology. They categorize the technologies regarding their relevance in particular customer journey stages, but they do not consider the features in a shopping app context. Zhao and Balagué (2015) structure app features according to companies' business goals such as product innovation or communication. However, this structure takes a retailer's point of view and neglects the consumer's perspective. Furthermore, they focus on branded mobile apps and not specifically on shopping apps.

Other studies find positive effects of mobile apps and (app) features on consumers' satisfaction and/or purchase intention (see e.g., Baier and Rese, 2020; Inman and Nikolova, 2017; Sinemus et al., 2022). Inman and Nikolova (2017) find positive effects of mobile apps and few features on consumers' purchase intention and word of mouth, but focus on grocery shopping. Sinemus et al. (2022) consider several shopping app features and find a significant impact of these features on customer satisfaction. However, they bundle app features regarding the customer journey stages and cross-channel options and do not analyze how single shopping app features contribute to satisfaction.

Studies further considered that customers may take some features for

Table 2
Research on outcomes of app features.

Authors	Method	Context	Outcome
Baier and Rese (2020)	Kano approach Cluster analysis	New physical store and online technologies	Satisfaction
Inman and Nikolova (2017)	ANOVA	Mobile apps as a whole and specific app feature (scan and go)	Purchase likelihood Satisfaction Word of mouth
Roggeveen and Sethuraman (2020)	Descriptive	Customer-interfacing retail technologies	–
Sinemus et al. (2022)	PLS	App features groups (pre-purchase, purchase, cross-channel)	Satisfaction
Zhao and Balagué (2015)	Descriptive	Mobile features used in branded apps	Retailer strategies
Zhao and Dholakia (2009)	Kano approach	Website attributes	Satisfaction
This study	**Kano approach** **Cluster analysis**	**25 app features**	**Satisfaction**

granted (must-be features), while they do not necessarily expect other features, but might be delighted if the retailer offers them (attractive features). Two studies applied the Kano et al. (1984) approach to categorize features according to their potential to create dissatisfaction if they are not offered and their potential to create satisfaction and delight if they are offered to customers. Zhao and Dholakia (2009) used the Kano et al. (1984) approach for categorizing selected website attributes. However, these do not cover the broad range of features offered in shopping apps today, as several new app features have been added in recent years. Baier and Rese (2020) used this approach to categorize physical store and online technologies. They consider some features that can also be included in a shopping app such as bonus cards, chat services, or click and collect. The authors further identify two segments. One segment categorizes most of the technologies as indifferent. The other segment categorizes more technologies as attractive such as reserve and collect, customization, and product testing. However, as mentioned before, the authors concentrate on the comparison of new online and physical store technologies and consider shopping apps as one technology.

In summary, previous literature leaves several research gaps. Firstly, several studies are either conceptual (Roggeveen and Sethuraman, 2020; Zhao and Balagué, 2015) or focus on selected app features and general feature groups (Sinemus et al., 2022) or specific product categories, such as groceries (Inman and Nikolova, 2017). Secondly, only two studies consider that features differently contribute to preventing dissatisfaction and delighting customers by application of the Kano et al. (1984) approach (Baier and Rese, 2020; Zhao and Dholakia, 2009). These studies focus on website attributes or general online and offline technologies, but not specifically on app features. However, knowledge which features consumers characterize as must-be, attractive, one-dimensional, or indifferent is highly relevant for designing apps. Thirdly, most previous research does not consider that customers differ in their expectations regarding shopping app features and consequently in how they categorize features in must-be, attractive, one-dimensional, or indifferent features. Some studies identified segments with a general focus on the mobile channel (Frasquet et al., 2019; Herhausen et al., 2019; Schneider and Zielke, 2020), but only Baier and Rese (2020) segmented customers based on their categorization of offline and online technologies along with the Kano et al. (1984) approach. This paper addresses these research gaps by categorizing a broad range of shopping app features with the Kano et al. (1984) approach. Subsequently, it segments customers based on their categorization pattern.

2.3. Theoretical framework

For analyzing how shopping app features contribute to customer satisfaction or dissatisfaction, we apply Kano's model of customer satisfaction (Kano et al., 1984). This approach is often used for the design of products or services, as it allows consideration of several features in a cost-saving manner (Baier and Rese, 2020). Kano's model has been used to categorize various objects such as different dimensions of price perception (Zielke, 2008), website attributes (Zhao and Dholakia, 2009), retailers' physical store services and technologies in the fast fashion context (Rese et al., 2019), outsourcing factors in the customer journey (Kranzbühler et al., 2019), service elements at a mega event (Chen et al., 2021), after-sales services in retailing (Shokouhyar et al., 2020), specific features of proactive services (Wenninger et al., 2022), hotel services (Kim et al., 2025), or attributes of mobile phones at various life cycle stages (Zhang et al., 2024). Hence, the Kano model has been applied in a variety of contexts. Some studies combine the Kano model with additional analyses to gain further insights. For example, Kim et al. (2025) and Wenninger et al. (2022) combine the Kano model with an importance performance analysis. Baier and Rese (2020) combined the Kano model with an additional cluster analysis for customer segmentation. We follow this approach by Baier and Rese (2020), as we aim to show which app features contribute to consumer (dis)satisfaction, and how this contribution differs between customer segments. The identification of different user clusters based on their feature categorization is important to develop implications for targeted feature advertising. Nevertheless, we consider also the importance of different features by using additional tests such as Berger et al.'s (1993) rule, where the sum of attractive, one-dimensional, and must-be feature categorizations indicates feature importance.

Kano's model traces back to Herzberg's two-factor theory of job satisfaction that distinguishes hygiene and motivator factors. While motivator factors increase job satisfaction, hygiene factors prevent job dissatisfaction (Herzberg, 1966; Herzberg et al., 1959). This implies asymmetric relationships between attribute performance and satisfaction. Kano et al. (1984) consider these asymmetric relationships in their model to categorize attributes or features depending on their contribution to satisfaction and their prevention of dissatisfaction.

- A: Attractive features. Their fulfillment leads to satisfaction, but non-fulfillment does not lead to dissatisfaction. Consequently, such features can delight customers, but they do not expect them.
- I: Indifferent features. These features are irrelevant for satisfaction formation. Consequently, fulfillment of these features does not matter.
- M: Must-be features. Their non-fulfillment leads to dissatisfaction, but their fulfillment does not lead to satisfaction. Consequently, consumers expect these features and take them for granted.
- O: One-dimensional features. These features have an impact on both satisfaction and dissatisfaction depending on their fulfillment.
- R: Reverse features. These features lead to satisfaction if they not fulfilled and dissatisfaction if they are fulfilled.
- Q: Questionable features. Consumers who do not follow one of the patterns above are categorized as questionable.

Kano et al. (1984) developed specific questions to categorize features into the abovementioned groups by asking respondents two questions for each feature:

Table 3
Kano's categorization table (adapted from Kano et al., 1984; Berger et al., 1993).

		Dysfunctional question				
		I like it that way	It must be that way	I am neutral	I can live with it that way	I dislike it that way
Functional question	I like it that way	Q	A	A	A	O
	It must be that way	R	I	I	I	M
	I am neutral	R	I	I	I	M
	I can live with it that way	R	I	I	I	M
	I dislike it that way	R	R	R	R	Q

- "If the feature is offered in the described way, how would you feel?" (functional question)
- "If the feature is not offered in the described way, how would you feel?" (dysfunctional question)

Respondents have to choose between five possible response options for each question. Features are then categorized based on the combination of answers to the two questions (see Table 3).

Categorizations are individual for each respondent. This means that one participant classifies a certain feature as attractive and another as indifferent. In order to summarize the results, Berger et al. (1993) propose the calculation of CS coefficients (customer satisfaction scores). The positive CS^+ score[1] expresses the contribution of a feature to satisfaction, while the negative CS^- score[2] indicates its contribution to dissatisfaction in the case that the feature is not offered. With the help of the CS scores it is possible to position the app features in a satisfaction-dissatisfaction grid (Appendix A).

3. Propositions

Retailers must decide how to design their shopping apps (see e.g., Boyd et al., 2019; Zhao and Balagué, 2015) and to integrate new arising app features (Al-Nabhani et al., 2021). Sinemus et al. (2022) show that retailers offer different features in their shopping apps. Some features are included in the majority of shopping apps, while other features are less common. Hence, retailers can categorize features in core features that are included in almost all shopping apps and supplementary features that are only offered in some apps.

As consumers tend to take frequently offered advantages for granted (see e.g., DelVecchio et al., 2007), we assume that consumers expect any app to have features that they are familiar with from many retailers (core features), while this is not the case for supplementary features. Therefore, core and supplementary features can contribute differently to satisfaction or prevention of dissatisfaction. As consumers are familiar with core features, negative consequences of their non-fulfillment should be higher than for supplementary features. Consequently, core features lead to higher negative customer satisfaction scores for non-fulfillment (CS^-) than supplementary features. As consumers are less familiar with supplementary features, they do not take them for granted and their fulfillment thus has higher potential to create satisfaction. Consequently, supplementary features lead to higher positive customer satisfaction scores for fulfillment (CS^+). Lo et al. (2016) also support this proposition by finding that standard features such as a store locator or shipment tracking lead to dissatisfaction if the performance of these factors falls below an acceptable level, while supplementary features such as bonus rewards or chat rooms are factors leading to satisfaction, if the performance reaches a certain level. Hence, we propose the following.

P1: Core features lead to higher negative customer satisfaction scores for non-fulfillment (CS^-) than supplementary features.

P2: Supplementary features lead to higher positive customer satisfaction scores for fulfillment (CS^+) than core features.

Consumers differ in their shopping motives and consequently in their shopping behavior. According to Babin et al. (1994) consumers can have a utilitarian or hedonic shopping orientation. Utilitarian shopping motives are mostly rational and the shopping is task-related, while hedonic shopping motives are more emotional and fun-related. Depending on the strength of such motives, customers differ in their buying or browsing orientation. Buyers, are more goal-directed and know what they want to buy (see e.g., Gilbride et al., 2015; Wang et al., 2011). We assume that buyers are more likely to expect particular features they need to achieve their purchase goals. Hence, they have a stronger tendency to categorize features as must-be. Browsers shop more spontaneous and search for inspiration (Beatty and Ferrell, 1998; Rook and Fisher, 1995) or want to kill their boredom (Sundström et al., 2019). Here, we assume that these consumers are more enthusiastic about app features and are more likely to categorize them as attractive features. Similarly, we also assume that app enjoyment and innovativeness increase the likelihood of categorizing features as attractive. Previous research has in particular identified innovativeness as an important profiling factor of segments related to channel use (see e.g., Nakano and Kondo, 2018). Based on these arguments, we expect that segments with a stronger tendency to categorize features as must be show a stronger buying orientation, while segments with a stronger tendency to categorize features as attractive show a stronger browsing orientation, app enjoyment, and innovativeness.

We further assume that customers who use apps only occasionally focus on basic functions and thus do not expect or value the diverse features. Similarly, as product involvement leads to the usage of more channels and touchpoints (see e.g., De Keyser et al., 2015; Herhausen et al., 2019), we assume that respondents with low product involvement are less interested in apps and their features and thus categorize many features as indifferent. Hence, we expect that segments with a stronger tendency to categorize features as indifferent have a lower usage frequency of apps and a lower product involvement than other segments. Based on the discussion above, we propose.

P3: Segments with a stronger tendency to categorize features as must-be show a stronger buying orientation than other segments.
P4: Segments with a stronger tendency to categorize features as attractive show a stronger (a) browsing orientation, (b) app enjoyment, and (c) innovativeness than other segments.
P5: Segments with a stronger tendency to categorize features as indifferent show a lower (a) app usage frequency and (b) product involvement than other segments.

4. Analysis and results

4.1. Pre-study

As our propositions distinguished between core and supplementary features, we conducted a pre-study to examine how frequently retailers offer the 25 features presented in Table 1 in their shopping apps. For this purpose, we examined 41 shopping apps from the largest sectors in the German e-commerce (EHI Retail Institute, 2021), fashion, electronics,

[1] $CS^+ = (A + O)/(M + O + A + I)$.
[2] $CS^- = (M + O)/(M + O + A + I)$.

furniture, and generalists. We defined core features as those that are offered by more than 80% of retailers and supplementary features as those offered by less than 80% (see Appendix B).

4.2. Methodology and sample

We design a questionnaire to categorize the 25 app features presented in Table 1 according to Kano's approach. For this purpose, we formulated a functional and a dysfunctional question for each feature (e.g., 'If the product availability check is offered in the described way, how would you feel?'; 'If the product availability check is not offered in the described way, how would you feel?') with the respective five answer options presented in Table 3. To ensure that all respondents had an idea of the respective feature, it was briefly explained as an introduction. We further measured some profiling factors (see Appendix C) and demographics (see Appendix D). We designed two versions of the questionnaire, one for fashion and one for electronics. Respondents were randomly assigned to the fashion or electronics questionnaire. We analyzed fashion and electronics, as these sectors generate the most revenue in e-commerce alongside generalists such as Amazon or Otto. In addition, retailers from these two sectors are most frequently represented in the top 100 e-commerce stores ranked by revenue (EHI Retail Institute, 2021). Hence, consumers are most likely to use shopping apps from retailers selling these two categories. We also included a filter question, asking which shopping channels (online, app, brick-and-mortar store) respondents have used within the last six months. In addition, we asked respondents which apps from a selection of 22 retailers from the sectors fashion, electronics, drug stores, and groceries (based on Germany's top-selling online stores) they have used. Based on these filter questions, we screened out respondents who have not used a shopping app in the respective product category. We excluded non-users of shopping apps in the respective categories to reduce biases, as app users differ from non-users in e.g., their loyalty toward the retailer (see Sinemus et al., 2022). Furthermore, non-users should be less familiar with the various app features and the features might be not valuable for them without using apps. When respondents indicated that they have used a shopping app of the randomly assigned product category within the last six months, we asked them to categorize the features according to Kano's model. As some features were relevant for only one of the two product categories (e.g., the size finder for fashion), we included 23 features for electronics and 24 features for fashion. We further included a quality question 'a full day has no more than 10 h' to screen out respondents who rush through the questionnaire and do not read questions and instructions thoroughly. Finally, we asked respondents for some socio-demographic information.

We collected data after pre-testing through a commercial online panel in Germany in September 2023. The panel provider invited panelists to complete the survey and closed the survey when quotas, which we explain below, were filled. Therefore, response rates are not available. All respondents used a fashion or electronics app within the last six months, i.e., we only included respondents who have used a shopping app in the randomly assigned product category in the last six months. Our sample is representative of the German population in terms of age (Statistisches Bundesamt, 2024). For this purpose, we divided the subjects into age cohorts (Baby Boomer, GenX, GenY, GenZ) in line with the German population, excluding those under 18 and over 67. We decided to include respondents from 18 to 67, as app users can be found in all age groups (Statista, 2024). Furthermore, we used quota for gender and the product category so that we have the same gender and product category distribution in all age groups. The online survey yielded 798 valid questionnaires. Respondents' average age is 45 (SD = 14.05) years and 49.9% are female and 50.1% are male. Most of the respondents live in a two-person household, are employed full-time and have an income of between EUR 2.000 and less than EUR 3.000. The majority of respondents (708) indicated that they have used all three channels (app, online shop, brick-and-mortar store) within the last six months. Only 32 respondents only used the physical store and the app, 48 respondents only used the online shop and the app and ten respondents only used the app and not any other channel.

For data analysis, we firstly categorized all features according to the scheme presented in Table 3 and calculated frequencies of individual categorizations and CS-coefficients (Berger et al., 1993). As classification is often ambiguous with respect to the most frequent categorization, we tested category strength (Lee and Newcomb, 1997) as an alternative to Fong's (1996) statistical significance test, following previous literature (see e.g., Zhao and Dholakia, 2009). The category strength is calculated from the difference between the shares of the two most frequent categories and must be higher than 6%, indicating a significant difference with a probability of 90% (Lee and Newcomb, 1997). Moreover, the total strength (Σ A, O, M) must be higher than 60%. As the threshold for the total strength has been arbitrarily determined (Zhao and Dholakia, 2009), we decided to follow Berger et al.'s (1993) rule, if the category strength indicates no significant difference between the two most frequent categories. Berger et al. (1993) propose the rule (A, O, M) <> (I, R, Q) that is applicable when one of the two most frequent categories is attractive, one-dimensional, or must be, while the other category is indifferent, reserved, or questionable. The rule states that if (Σ A, O, M) > (Σ I, R, Q), the feature should be classified as attractive, one-dimensional, or must be, depending on the frequency, and if (Σ I, R, Q) > (Σ A, O, M), the feature should be classified as indifferent, reversed, or questionable. The classification is based on the most frequently mentioned category. If the two most frequently categorizations are in the same group (e.g., attractive and one-dimensional), the (A, O, M) < > (I, R, Q) rule is not applicable and this item has to categorized as mixed mode.

In a second step, we conducted a two-step cluster analysis (Chiu et al., 2001) to identify customer segments differing in their categorization pattern and characterized these segments by profiling factors. The two-step clustering approach allows missing values and is able to select the most appropriate cluster number based on the Bayesian Information Criterion (BIC) (Bauer et al., 2020). We clustered respondents based on their individual tendencies to categorize features as must-be, attractive, one-dimensional, and indifferent (how many features they characterized respectively). Afterwards we conducted MANOVA with either Tukey or Games-Howell post-hoc tests (depending on variance homogeneity) to analyze differences between the segments.

4.3. Main results

Table 4 shows how frequently respondents categorized the 25 features as must-be, attractive, one-dimensional, indifferent, questionable, and reverse features along with the most frequent categorization and the CS-coefficients. Based on the most frequent categorization, many features are indifferent. In particular, the online magazine, click and meet, chatbot, augmented reality, inbox, personal product recommendations, sharing product links, live chat, and get the look were categorized as indifferent by more than half of the respondents. Personal product recommendations, inbox, and get the look were even rated as reverse by more than 10 percent of respondents. Except sharing product links, all aforementioned features are supplementary features. Four features were most frequently characterized as must-be features: product availability check, product finding tools, shipment tracking, and wish list, which are all core features. Also, four features were most frequently characterized as attractive features: comparison list, loyalty program, re-availability e-mail, and size finder, which are all supplementary features. Based on CS-coefficients, the supplementary features visual search and check and reserve can also be considered as attractive features. In general, the negative CS scores tend to be higher for core features and the positive CS scores tend to be higher for supplementary features. This is in line with our propositions of P1 and P2.

We further presented the categorization in Table 4 by calculating category strength. As results on Table 5 show, category strength is

Table 4
Results of Kano model.

	A	I	M	O	Q	R	Mode	CS^+	CS^-
Core features									
Data preservation	75	**390**	202	61	13	57	I	0.1868	0.3613
Electronic receipt	270	**312**	84	86	26	20	I	0.4734	0.2261
Product availability check	193	202	**227**	150	21	5	M	0.4443	0.4883
Product finding tools	138	234	**284**	117	15	10	M	0.3299	0.5188
Sharing product links	149	**416**	134	55	17	27	I	0.2706	0.2507
Shipment tracking	127	135	**347**	165	16	8	M	0.3773	0.6615
Store locator	186	**284**	211	89	20	8	I	0.3571	0.3896
Wish list	150	194	**288**	131	25	10	M	0.3683	0.5491
Supplementary features									
Augmented reality	248	**439**	7	44	20	40	I	0.3957	0.0691
Chatbot	159	**462**	47	42	19	69	I	0.2831	0.1254
Check and reserve	300	**311**	67	93	19	8	I	0.5097	0.2075
Click and collect	250	**361**	59	95	23	10	I	0.4510	0.2013
Click and meet	195	**493**	25	44	21	20	I	0.3157	0.0911
Comparison list	**208**	72	46	65	7	0	A	0.6982	0.2839
Get the look	82	**205**	6	17	7	83	I	0.3194	0.0742
Inbox	146	**435**	47	41	19	110	I	0.2795	0.1315
Live chat	233	**410**	51	40	22	42	I	0.3719	0.1240
Loyalty program	**320**	281	58	105	21	13	A	0.5563	0.2134
Online magazine	116	**569**	15	25	17	56	I	0.1945	0.0552
Payment via app	201	**352**	124	68	19	34	I	0.3611	0.2577
Personal product recommendations	180	**432**	35	30	19	102	I	0.3102	0.0960
QR code scanner	292	**374**	39	56	23	14	I	0.4573	0.1248
Re-availability e-mail	**292**	204	140	125	23	14	A	0.5480	0.3482
Size finder	**169**	100	48	58	12	13	A	0.6053	0.2827
Visual search	336	**356**	20	56	17	13	I	0.5104	0.0990

Note: bold = most frequent classification.

Table 5
Results of Kano model's quality test.

	Category						Mode	Category strength	A + O + M	I + R + Q	Final mode
	A[%]	I[%]	M[%]	O[%]	Q[%]	R[%]					
Core features											
Data preservation	9.40	**48.87**	**25.31**	7.64	1.63	7.14	I	23.56*	42.36	57.64	I
Electronic receipt	**33.83**	**39.10**	10.53	10.78	3.26	2.51	I	5.26[a]	55.14	44.86	A[b]
Product availability check	24.19	**25.31**	**28.45**	18.80	2.63	0.63	M	3.13[a]	71.43	28.57	M
Product finding tools	17.29	**29.32**	**35.59**	14.66	1.88	1.25	M	6.27*	67.54	32.46	M
Sharing product links	**18.67**	**52.13**	16.79	6.89	2.13	3.38	I	33.46*	42.36	57.64	I
Shipment tracking	15.91	16.92	**43.48**	**20.68**	2.01	1.00	M	22.81*	80.08	19.92	M
Store locator	23.31	**35.59**	**26.44**	11.15	2.51	1.00	I	9.15*	60.90	39.10	I
Wish list	18.80	**24.31**	**36.09**	16.42	3.13	1.25	M	11.78*	71.30	28.70	M
Supplementary features											
Augmented reality	**31.08**	**55.01**	0.88	5.51	2.51	5.01	I	23.93*	37.47	62.53	I
Chatbot	**19.92**	**57.89**	5.89	5.26	2.38	8.65	I	37.97*	31.08	68.92	I
Check and reserve	**37.59**	**38.97**	8.40	11.65	2.38	1.00	I	1.38[a]	57.64	42.36	A[b]
Click and collect	**31.33**	**45.24**	7.39	11.90	2.88	1.25	I	13.91*	50.63	49.37	I
Click and meet	**24.44**	**61.78**	3.13	5.51	2.63	2.51	I	37.34*	33.08	66.92	I
Comparison list	**52.26**	**18.09**	11.56	16.33	1.76	0.00	A	34.17*	80.15	19.85	A
Get the look	**20.50**	**51.25**	1.50	4.25	1.75	20.75	I	30.75*	26.25	73.75	I
Inbox	**18.30**	**54.51**	5.89	5.14	2.38	13.78	I	36.22*	29.32	70.68	I
Live chat	**29.20**	**51.38**	6.39	5.01	2.76	5.26	I	22.18*	40.60	59.40	I
Loyalty program	**40.10**	**35.21**	7.27	13.16	2.63	1.63	A	4.89[a]	60.53	39.47	A
Online magazine	**14.54**	**71.30**	1.88	3.13	2.13	7.02	I	56.77*	19.55	80.45	I
Payment via app	**25.19**	**44.11**	15.54	8.52	2.38	4.26	I	18.92*	49.25	50.75	I
Personal product recommendations	**22.56**	**54.14**	4.39	3.76	2.38	12.78	I	31.58*	30.70	69.30	I
QR code scanner	**36.59**	**46.87**	4.89	7.02	2.88	1.75	I	10.28*	48.50	51.50	I
Re-availability e-mail	**36.59**	**25.56**	17.54	15.66	2.88	1.75	A	11.03*	69.80	30.20	A
Size finder	**42.25**	**25.00**	12.00	14.50	3.00	3.25	A	17.25*	68.75	31.25	A
Visual search	**42.11**	**44.61**	2.51	7.02	2.13	1.63	I	2.51[a]	51.63	48.37	A[b]

Notes: bold = most and second most frequent classification; [a] (O + A + M) < > (I + R + Q) rule; [b] = change in category mode; * = Categorization significant at a ten-percent level according to Lee and Newcomb (1997).

higher than 6% for all features except three: check and reserve, electronic receipt, and visual search. For these three features the two most frequently chosen categorizations are A and I. Therefore the (A, O, M) < > (I, R, Q) rule is applicable. As for all three features the sum of attractive, one-dimensional, and must-be categorizations is higher than the sum of indifferent, reserve, and questionable categorizations (Σ A, O, M) > (Σ I, R, Q), we adapted the final mode from indifferent to attractive. These results are also in line with the results of the CS-coefficients regarding visual search and check and reserve. Based on the sum of attractive, one-dimensional, and must-be features (Σ A, O, M), comparison list, shipment tracking, product availability checks, and wish list are the most important features, followed by re-availability e-mail, size

finder, and product finding tools.

In an additional analysis, we explored differences in categorizations between the sectors fashion and electronics. Results show that most features are categorized in the same mode, except a few differences. Fashion respondents categorize the electronic receipt as indifferent instead of attractive. Electronic respondents categorize the store locator as attractive instead of indifferent. Furthermore, the QR code scanner is categorized as mixed mode (see Appendices E to H).

4.4. Cluster analysis

The two-step cluster analysis revealed a three-cluster solution. The first cluster is the largest with 385 respondents, cluster 2 covers 142, and cluster 3 covers 270 individuals. Table 6 shows that the three clusters have different tendencies to categorize features as must-be, attractive, one-dimensional, and indifferent. Cluster one has the largest tendency to categorize features as must-be. Cluster two categorizes more features as attractive and one-dimensional, while cluster 3 has a strong tendency to categorize features as indifferent.

Figs. 1–3 further categorize features for the three clusters based on CS scores. The grid categorizes features as attractive ($CS^+ > .5/CS^- < .5$), indifferent ($CS^+ < .5/CS^- < .5$), must-be ($CS^+ < .5/CS^- > .5$), and one-dimensional ($CS^+ > .5/CS^- > .5$). Cluster 1 characterizes 11 features as indifferent, 9 features as attractive and 5 features as must-be. All must-be features are core features. Cluster 2 characterizes 2 features as indifferent, 19 features as attractive and 4 features as one-dimensional. All one-dimensional features are core features. Cluster 3 characterizes all features except one (comparison list) as indifferent. However, it can be seen that core features show higher CS^- coefficients than supplementary features. It is further interesting that some feature groups are positioned close to each other in all three segments. Product availability check, product finding tools, shipment tracking, and wish list (all core features) are the only must-be requirements in segment 1, the only one-dimensional requirements in segment 2 and show the highest CS^- coefficients in cluster 3. A comparison list shows the highest CS^+ coefficient in all clusters and is positioned close to loyalty program, re-availability e-mail, and size finder.

We further analyzed how the segments differ in several profiling factors according to propositions 3 to 5 (see Table 7 and further demographics in Appendix D). We find no support for P3, as segment 1 with most must-be features does not show a significantly higher buying orientation than the other segments. However, we find some support for P4. Segment 2 with the largest tendency to categorize features as attractive shows the highest values for browsing tendency, app enjoyment, and innovativeness. In particular browsing orientation is significantly stronger than in segment 1, app enjoyment is significantly higher than in segment 3, and innovativeness is significantly higher compared to both other segments. Regarding P5 we find support as the indifferent segment 3 shows the lowest usage frequency of shopping apps along with a lower product involvement than clusters 1 and 2. Hence, we find a conservative core feature segment, an inspiration and innovation seeking attractive segment, and an uninvolved indifferent segment. We further checked if the segments differ in gender and age. While a Chi^2 test with Bonferroni correction did not indicate gender differences, segments differ in age with significantly younger respondents in segment 1.

As we collected data for two product groups (fashion and electronics), we checked if product groups influenced cluster membership. Results show a balanced distribution of fashion and electronic app users for all three clusters. Segment 1 contains 48.1%, segment 2 50%, and segment 3 53% fashion app users. Furthermore, a Chi^2 test (1.531; $p = .467$) shows that the cluster distribution does not differ between both sectors.

5. Discussion

The results show that app features contribute differently to preventing dissatisfaction and increasing satisfaction. While all must-be features are core features that are offered by almost all retailers, all attractive features are supplementary features offered in only some shopping apps. Surprisingly, many features were categorized as indifferent. This indicates that consumers do not care about many features. While our distinction in core and supplementary features helps to understand feature categorization, we did not observe such pattern for alternative feature groupings (e.g., Sinemus et al., 2022). For example, we find among the attractive features pre-purchase features (comparison list), transaction features (loyalty program), and cross-channel features (check and reserve). Among the must-be features, we also find pre-purchase features (product finding tools), transaction features (shipment tracking), and cross-channel features (availability check).

Furthermore, the categorization is mostly equal for the fashion and electronics sector with a few exceptions. Consumers perceive the electronic receipt as indifferent for fashion, but attractive for electronics (based on the (A, O, M) < > (I, R, Q) rule). This could be as the electronic receipt is more relevant for electronics in the case of a warranty claim. Consumers from the electronics scenario are further more inclined to perceive the QR code scanner attractive, as the QR code scanner directs them to the online store and thus provides further information about the product. Electronic products are more utilitarian (Blom et al., 2021) and information-intensive. This could also be the reason why consumers in

Table 6
Characterizing factors of segmentation.

	Segments			Total	F value (p)	Results of post-hoc tests
	1	2	3			
N	385	142	270	797		
A	32.20%	**40.62%**	11.86%	26.81%	229.412 (<.001)	2 > 1***
						2 > 3***
						1 > 3***
I	34.04%	21.75%	**66.37%**	42.80%	698.495 (<.001)	3 > 1***
						3 > 2***
						1 > 2***
M	**22.27%**	2.83%	7.66%	13.86%	275.786 (<.001)	1 > 2***
						1 > 3***
						3 > 2***
O	7.42%	**27.39%**	4.25%	9.90%	194.386 (<.001)	2 > 1***
						2 > 3***
						1 > 3***

Notes: A = attractive; I = indifferent; M = must-be; O = one-dimensional; *** = $p < .001$; the percentage values don't add up to 100% as the questionable and reverse categorizations are not included.

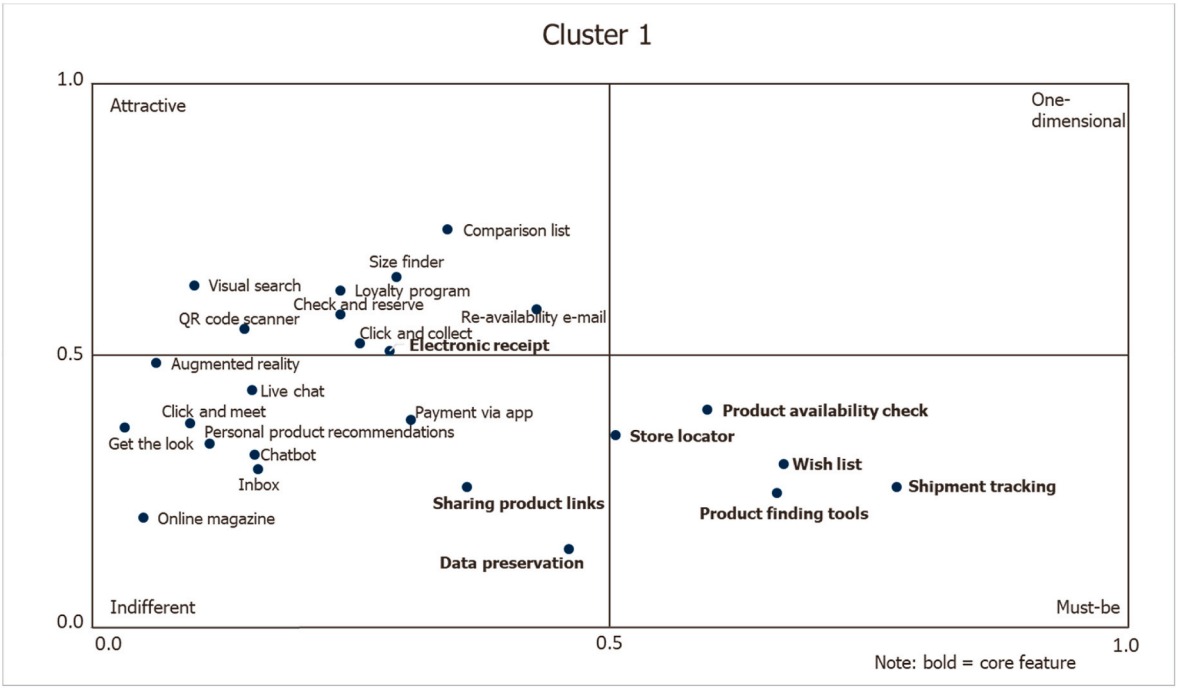


Fig. 1. Satisfaction-dissatisfaction-grid cluster 1.

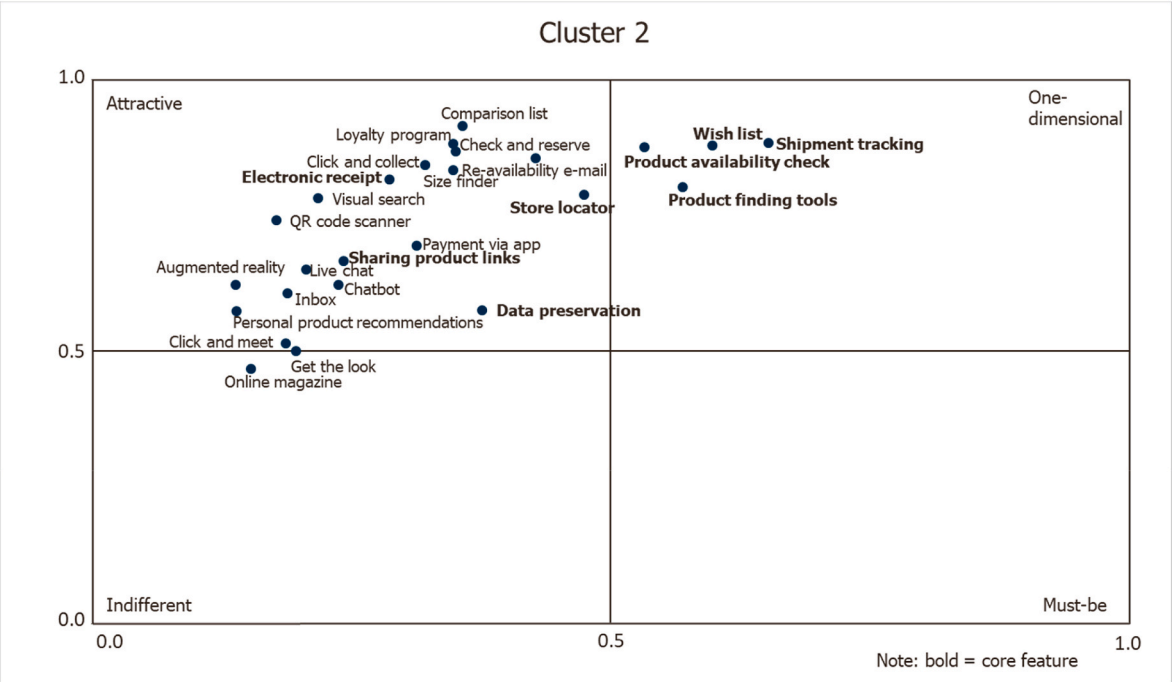


Fig. 2. Satisfaction-dissatisfaction-grid cluster 2.

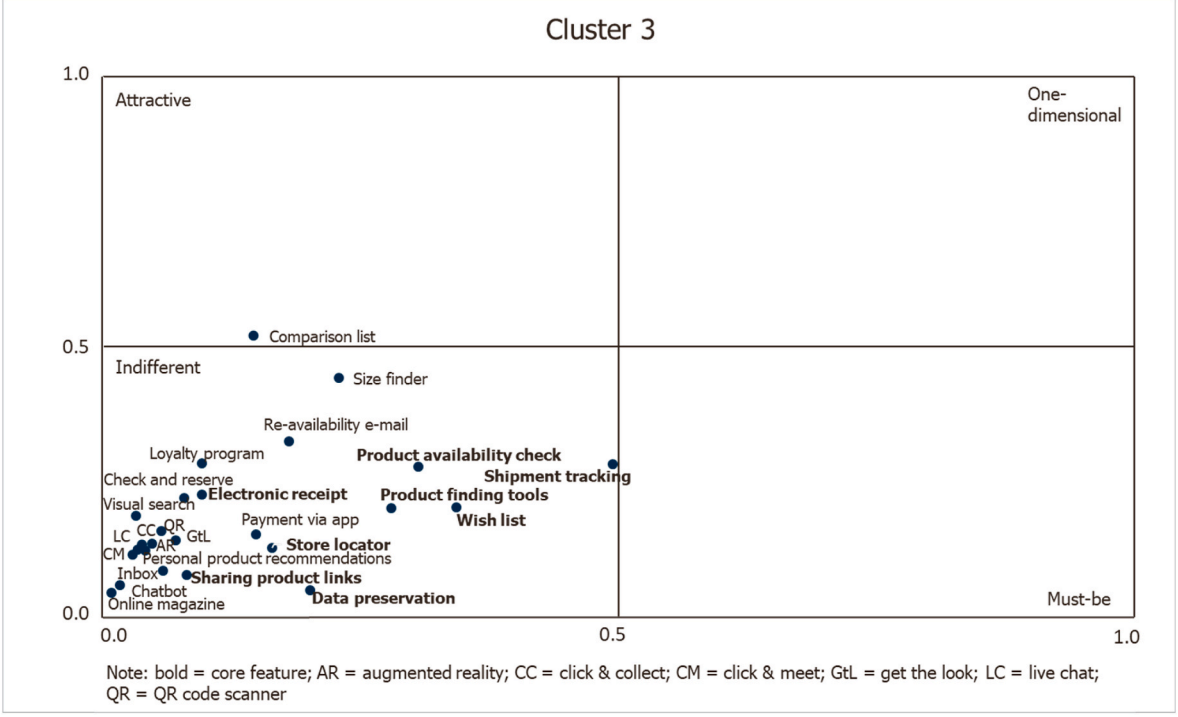


Fig. 3. Satisfaction-dissatisfaction-grid cluster 3.

Table 7
Characterization of shopping app feature segments - profiling factors.

	Segment			Total	F value (p)	Results of post-hoc tests	N
	1	2	3				
N	385	142	270	779			
Buying orientation	5.71	5.85	5.59	5.69	1.746 (.175)	–	779
Browsing orientation	3.44	3.94	3.62	3.59	4.865 (.008)	1 < 2**	779
						1 < 3*	
Usage frequency	3.25	3.32	2.87	3.14	6.889 (.001)	1 > 3**	779
						2 > 3**	
App enjoyment	5.37	5.63	5.02	5.30	7.466 (<.001)	1 > 3**	779
						2 > 3**	
Product involvement	5.36	5.43	5.07	5.28	3.463 (.032)	1 > 3**	779
						2 > 3**	
Innovativeness	3.55	3.88	3.62	3.63	2.865 (.058)	1 < 2**	779
						3 < 2*	
Gender	192 f	85 f	120 f	397 f	$Chi^2 = 8.846$ (.012)	–[c]	797
	193 m	57 m	150 m	400 m			
Age	43.09	46.45	46.52	44.85	5.931 (.003)	1 < 2**	797
						1 < 3**	

Notes: **p < .05; *p < .10; covariates: age, gender; f = female, m = male; [c] = based on Bonferroni correction.

the electronic scenario perceive the store locator as attractive and not indifferent.

The segmentation revealed three segments. The first segment covers respondents with a higher tendency of must-be categorizations, in particular for core features. Against our expectations, this segment does not show a higher buying orientation than other segments. As all segments show a similar and high buying orientation, this could indicate that a high buying orientation is a strong driver of app usage in general, while the browsing orientation is more relevant for feature categorization. The second segment with a higher tendency of must-be categorizations shows the highest browsing orientation, app enjoyment, and innovativeness. In comparison, the first must-be segment has a significantly lower browsing orientation and innovativeness. Therefore, we can describe segment 1 as a conservative core feature segment and segment 2 as an innovative inspiration seeking segment. Finally, segment 3 is mostly indifferent and as expected characterized by low

usage frequency and low product involvement.

5.1. Management implications

The results help retailers to prioritize shopping app features. Firstly, retailers must include must-be features (Berger et al., 1993). In particular the core features were categorized as must-be requirements for the overall sample and they also show the highest CS^- coefficients in all segments. Hence, shopping apps should at least include product finding tools, a wish list, shipment tracking, and product availability checks. Secondly, retailers should selectively decide to include features that are categorized as attractive. Features, such as comparison list, loyalty program, re-availability e-mail, size finder, visual search, electronic receipt, or check and reserve are attractive features for the overall sample and in both segments 1 and 2. Other features, such as augmented reality, live chat, or payment via app are only attractive features in segment 2. Hence, retailers should preferably offer these features when their target group is innovative and searches for inspiration. However, when they target customers with low product involvement they may not offer an app or limit it to basic shopping functions, as customers do not expect or value any features. In the worst-case consumers could be overloaded and get confused. Some features, such as product recommendations, inbox, and get the look were even categorized as reverse by several respondents and therefore should be carefully used.

In addition, retailers should consider which products they offer. For example, detailed product information and product comparisons are particularly important for electronics. Therefore, retailers offering electronics or other information-intensive products should consider app features that help consumers finding the information they need, such as a QR code scanner or a store locator. Furthermore, total strength (Σ A, O, M) shows that a comparison list is very important for consumers as more than three-quarters indicate it as attractive, must-be, or one-dimensional feature. Regarding the fashion sector, a size finder is very important to consumers, with two thirds categorizing it as an important feature. For retailers, the size finder can also have the advantage of reducing the return rate, as consumers can find the right size immediately without having to order different sizes.

In addition, retailers should constantly monitor the market to distinguish between core and supplementary features. Supplementary features can evolve over time into core features (see e.g., Zhao and Dholakia, 2009), which, as mentioned above, should be offered in an app to prevent dissatisfaction. In addition, retailers can differentiate themselves from competitors with supplementary features, which increasingly contribute to satisfaction. It is therefore important to identify new developments in app features, for example new features using AI to support the shopping process.

The results have further implications for communicating feature benefits. For example, retailers can advertise feature benefits toward indifferent customer segments. Furthermore, retailers can structure features according to their categorization in the app design, so that customers with a focus on core features are not distracted by the supplementary features.

5.2. Theoretical contribution

Our study extends the existing literature in multiple ways. Firstly, while many previous studies analyzed how shopping apps as a whole or groups of features influence outcomes such as satisfaction or loyalty (Sinemus et al., 2022), we analyze 25 app features separately. Secondly, by applying the Kano model, we considered that app features play a different role in preventing dissatisfaction and increasing satisfaction. Our study thus contributes to research applying the Kano model in similar but different contexts (Baier and Rese, 2020; Zhao and Dholakia, 2009). A separate analysis of app features is particularly relevant, as the role of features may differ from websites because of the more convenient access to an app and personalized content, and as some features are not or less frequently offered on websites. Thirdly, we show that penetration of shopping apps in practice (core and supplementary features) influences the categorization of features as must-be (core features) and attractive (supplementary features). As we observed no clear pattern of feature categorization for alternative feature groupings (e.g., pre-purchase, transaction, and cross-channel features; see Sinemus et al., 2022) our study suggests an alternative grouping criterion that is relevant for understanding how features contribute to satisfaction or dissatisfaction. Fourthly, our study contributes to understanding individual differences in the categorization of features. We identified three segments with tendencies to categorize features as must-be, attractive, and indifferent. In addition, browsing orientation, app enjoyment, innovativeness, app usage frequency, and product involvement differ between the segments and explain categorization patterns.

5.3. Limitations and future research

Our study has some limitations and provides directions for future research. The first limitation refers to Kano's model itself. The model does not measure satisfaction directly, it only assumes that the answer categories of the functional and the dysfunction question indicate how features contribute to satisfaction or dissatisfaction. Furthermore, the model does not define the object of satisfaction. In our context it could be the satisfaction with shopping in general, a shopping trip or the satisfaction with the shopping app. Future research could therefore supplement our results by analyzing relationships between app features and satisfaction more directly. Nevertheless, Sauerwein (1999) found empirical support for the validity of Kano's model and its relation to customer satisfaction.

Moreover, as we found that the penetration of app features influences their categorization, future research should analyze new features at different points in time. New features may be first attractive features, but are later taken for granted. Related to this, our list of 25 features covers a broad range of features but such a list can never be complete, as new features arise and may replace old features. Future research should therefore update our feature list. Future research may also consider that the range of features and their design may differ between retail sectors. Single features can further differ in their design and implementation. For example, implementations of check and reserve may differ in their display of inventory, reservation confirmation, and pick-up time (Ortlinghaus and Zielke, 2019). Future research can therefore also analyze customer expectations regarding the design of single features. Finally, we segmented customers based on their feature categorization, but future research may also compare customer segments based on which channels customers use in addition to the app (store, online shop, catalog, etc.).

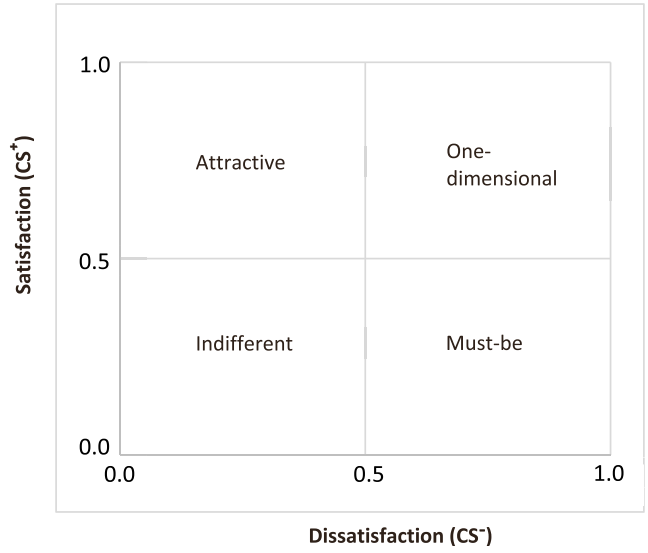
CRediT authorship contribution statement

Kathrin Sinemus: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stephan Zielke:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Thomas Dobbelstein:** Writing – review & editing, Funding acquisition.

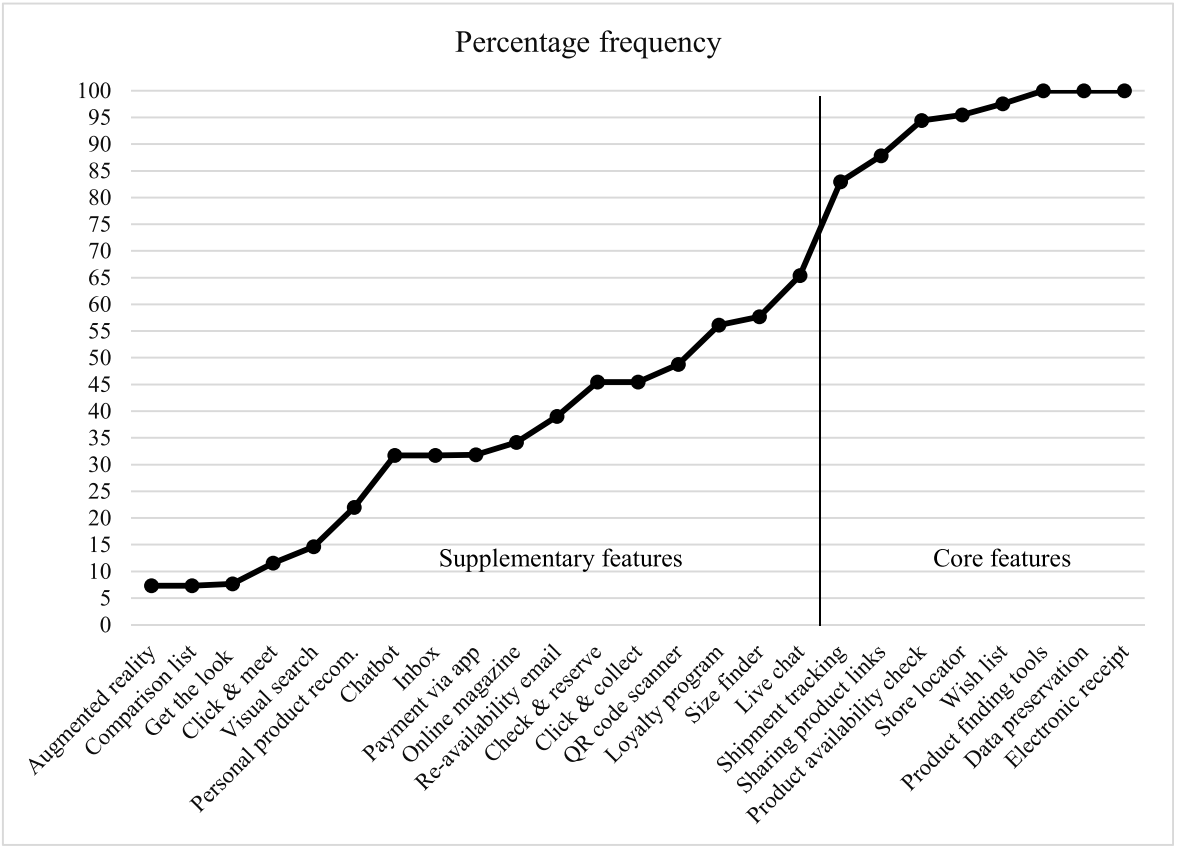
Declaration of competing interest

None.

Appendix A. Satisfaction-dissatisfaction-grid



Appendix B. Categorization of core and supplementary features



Appendix C. Profiling characteristic measurement

Variable	Item	Authors
Buying orientation	I usually use shopping apps to buy something specific.	Adapted from Büttner et al. (2014)
Browsing orientation	I usually use shopping apps without any prior concrete intention to buy.	Adapted from Büttner et al. (2014)
Usage frequency of shopping apps	How often do you use shopping apps from the apparel/electronic sector?	–
App enjoyment	Using shopping apps is enjoyable.	Venkatesh et al. (2012)
Product involvement	Overall, I am someone who cares about what items he/she buys.	De Wulf et al. (2001)
Innovativeness	I always use the newest products.	Adapted by Konuş et al. (2008)

Appendix D. Demographic data of the segments.

Variable	Conservative core feature segment (n = 385)		Inspiration and innovation seeking attractive segment (n = 142)		Uninvolved indifferent segment (n = 270)	
	n	%	n	%	n	%
Gender						
Female	192	49.9	85	59.9	120	44.4
Male	193	50.1	57	40.1	150	55.6
Age						
Ø	43		47		47	
Median	42		49		49	
(σ)	(14.35)		(13.78)		(13.51)	
Professional stage						
Full-time	209	54.3	73	51.4	151	56.1
Part-time	57	14.8	28	19.7	35	13.0
Student/Apprenticeship/Pupil	45	11.7	10	7.0	17	6.3
Pensioner	38	9.9	15	10.6	35	13.0
Other	35	9.1	16	11.2	31	11.6
No answer	1	0.3	–	–	1	0.4
Household Income						
<1000 EUR	23	6.0	6	4.2	19	7.1
1000 < 2000 EUR	52	13.5	26	18.3	56	20.4
2000 < 3000 EUR	87	22.6	35	24.7	49	18.1
3000 < 4000 EUR	57	14.8	18	12.6	46	17.0
4000 < 5000 EUR	64	16.6	18	13.6	38	14.1
5000 < 6000 EUR	38	9.9	11	7.7	24	8.9
>6000 EUR	34	8.8	12	8.4	13	4.9
No answer	30	7.8	16	11.3	25	9.3
Household size (adults and children)						
1 person	106	27.6	33	23.2	79	28.1
2 persons	150	39.0	56	39.4	97	35.9
3 persons	62	16.1	28	19.7	51	18.9
4 persons	41	10.6	24	16.9	32	11.9
>4 persons	26	6.7	1	0.7	14	5.2

Appendix E. Results of Kano model (fashion).

	A	I	M	O	Q	R	Mode	CS^+	CS^-
Core features									
Data preservation	36	**202**	95	32	6	29	I	0.1863	0.3479
Electronic receipt	130	**157**	45	42	15	11	I	0.4599	0.2326
Product availability check	100	103	**109**	76	12	0	M	0.4536	0.4768
Product finding tools	70	119	**136**	58	7	10	M	0.3342	0.5065
Sharing product links	66	**231**	55	26	6	16	I	0.2434	0.2143
Shipment tracking	59	62	**183**	83	8	5	M	0.3669	0.6873
Store locator	88	**154**	104	42	7	5	I	0.3351	0.3763
Wish list	65	96	**142**	81	9	7	M	0.3802	0.5807
Supplementary features									
Augmented reality	125	**214**	3	25	10	23	I	0.4087	0.0763
Chatbot	71	**235**	27	23	10	34	I	0.2640	0.1404
Check and reserve	150	**152**	29	53	10	6	I	0.5286	0.2135
Click and collect	119	**178**	23	59	12	9	I	0.4697	0.2164
Click and meet	74	**268**	13	19	10	16	I	0.2487	0.0856

(*continued on next page*)

(*continued*)

	A	I	M	O	Q	R	Mode	CS^+	CS^-
Comparison list	–	–	–	–	–	–	–	–	–
Get the look	82	**205**	6	17	7	83	I	0.3194	0.0742
Inbox	66	**216**	25	21	8	64	I	0.2652	0.1402
Live chat	100	**220**	27	18	12	23	I	0.3233	0.1233
Loyalty program	**140**	136	34	70	11	9	A	0.5526	0.2737
Online magazine	59	**291**	9	14	4	23	I	0.1957	0.0617
Payment via app	104	**178**	57	32	11	18	I	0.3666	0.2399
Personal product recommendations	104	**202**	22	20	8	44	I	0.3563	0.1207
QR code scanner	134	**196**	22	32	8	8	I	0.4323	0.1406
Re-availability e-mail	**131**	101	79	68	12	9	A	0.5251	0.3879
Size finder	**169**	100	48	58	12	13	A	0.6053	0.2827
Visual search	169	**175**	9	32	7	8	I	0.5221	0.1065

Note: bold = most frequent classification.

Appendix F. Results of Kano model's quality test (fashion).

	Category						Mode	Category strength	A+O+M	I+R+Q	Final mode
	A[%]	I[%]	M[%]	O[%]	Q[%]	R[%]					
Core features											
Data preservation	9.00	**50.50**	23.75	8.00	1.50	7.25	I	26.75*	40.75	59.25	I
Electronic receipt	32.50	**39.25**	11.25	10.50	3.75	2.75	I	6.75*	54.25	45.75	I
Product availability check	25.00	25.75	**27.25**	19.00	3.00	0.00	M	1.50[a]	71.25	28.75	M
Product finding tools	17.50	29.75	**34.00**	14.50	1.75	2.50	M	4.25[a]	66.00	34.00	M
Sharing product links	16.50	**57.75**	13.75	6.50	1.50	4.00	I	41.25*	36.75	63.25	I
Shipment tracking	14.75	15.50	**45.75**	20.75	2.00	1.25	M	25.00*	81.25	18.75	M
Store locator	22.00	**38.50**	26.00	10.50	1.75	1.25	I	12.50*	58.50	41.50	I
Wish list	16.25	24.00	**35.50**	20.25	2.25	1.75	M	11.50*	72.00	28.00	M
Supplementary features											
Augmented reality	31.25	**53.50**	0.75	6.25	2.50	5.75	I	22.25*	38.25	61.75	I
Chatbot	17.75	**58.75**	6.75	5.75	2.50	8.50	I	41.00*	30.25	69.75	I
Check and reserve	37.50	**38.00**	7.25	13.25	2.50	1.50	I	0.50[a]	58.00	42.00	A[b]
Click and collect	29.75	**44.50**	5.75	14.75	3.00	2.25	I	14.75*	50.25	49.75	I
Click and meet	18.50	**67.00**	3.25	4.75	2.50	4.00	I	48.50*	26.50	73.50	I
Comparison list	–	–	–	–	–	–	–	–	–	–	–
Get the look	20.50	**51.25**	1.50	4.25	1.75	20.75	I	30.75*	26.25	73.75	I
Inbox	16.50	**54.00**	6.25	5.25	2.00	16.00	I	37.50*	28.00	72.00	I
Live chat	25.00	**55.00**	6.75	4.50	3.00	5.75	I	30.00*	36.25	63.75	I
Loyalty program	**35.00**	34.00	8.50	17.50	2.75	2.25	A	1.00[a]	61.00	39.00	A
Online magazine	14.75	**72.75**	2.25	3.50	1.00	5.75	I	58.00*	20.50	79.50	I
Payment via app	26.00	**44.50**	14.25	8.00	2.75	4.50	I	18.50*	48.25	51.75	I
Personal product recommendations	26.00	**50.50**	5.50	5.00	2.00	11.00	I	24.50*	36.50	63.50	I
QR code scanner	33.50	**49.00**	5.50	8.00	2.00	2.00	I	15.50*	47.00	53.00	I
Re-availability e-mail	**32.75**	25.25	19.75	17.00	3.00	2.25	A	7.50[a]	69.50	30.50	A
Size finder	**42.25**	25.00	12.00	14.50	3.00	3.25	A	17.25*	68.75	31.25	A
Visual search	42.25	**43.75**	2.25	8.00	1.75	2.00	I	1.50[a]	52.50	47.50	A[b]

Notes: bold = most and second most frequent classification; [a] (O + A + M) < > (I + R + Q) rule; [b] = change in category mode; * = categorization significant at a ten-percent level according to Lee and Newcomb (1997).

Appendix G. Results of Kano model (electronics).

	A	I	M	O	Q	R	Mode	CS^+	CS^-
Core features									
Data preservation	39	**188**	107	29	7	28	I	0.1873	0.3747
Electronic receipt	140	**155**	39	44	11	9	I	0.4868	0.2196
Product availability check	93	99	**118**	74	9	5	M	0.4349	0.5000
Product finding tools	68	115	**148**	59	8	0	M	0.3256	0.5308
Sharing product links	83	**185**	79	29	11	11	I	0.2979	0.2872
Shipment tracking	68	73	**164**	82	8	3	M	0.3876	0.6357
Store locator	98	**130**	107	47	13	3	I	0.3796	0.4031
Wish list	85	98	**146**	50	16	3	M	0.3562	0.5172
Supplementary features									
Augmented reality	123	**225**	4	19	10	17	I	0.3827	0.0620
Chatbot	88	**227**	20	19	9	35	I	0.3023	0.1102
Check and reserve	150	**159**	38	40	9	2	I	0.4910	0.2016
Click and collect	131	**183**	36	36	11	1	I	0.4326	0.1865
Click and meet	121	**225**	12	25	11	4	I	0.3812	0.0966
Comparison list	**208**	72	46	65	7	0	A	0.6982	0.2839
Get the look	–	–	–	–	–	–	–	–	–
Inbox	80	**219**	22	20	11	46	I	0.2933	0.1232
Live chat	133	**190**	24	22	10	19	I	0.4201	0.1247
Loyalty program	**180**	145	24	35	10	4	A	0.5599	0.1536

(*continued on next page*)

(*continued*)

	A	I	M	O	Q	R	Mode	CS^+	CS^-
Online magazine	57	**278**	6	11	13	33	I	0.1932	0.0483
Payment via app	97	**174**	67	36	8	16	I	0.3556	0.2754
Personal product recommendations	76	**230**	13	10	11	58	I	0.2614	0.0699
QR code scanner	158	**178**	17	24	15	6	I	0.4828	0.1088
Re-availability e-mail	**161**	103	61	57	11	5	A	0.5707	0.3089
Size finder	–	–	–	–	–	–	–	–	–
Visual search	167	**181**	11	24	10	5	I	0.4987	0.0914

Note: bold = most frequent classification.

Appendix H. Results of Kano model's quality test (electronics).

	Category						Mode	Category strength	A + O + M	I + R + Q	Final mode
	A[%]	I[%]	M[%]	O[%]	Q[%]	R[%]					
Core features											
Data preservation	9.80	**47.24**	**26.88**	7.29	1.76	7.04	I	20.35*	43.97	56.03	I
Electronic receipt	**35.18**	**38.94**	9.80	11.06	2.76	2.26	I	3.77[a]	56.03	43.97	A[b]
Product availability check	23.37	**24.87**	**29.65**	18.59	2.26	1.26	M	4.77[a]	71.61	28.39	M
Product finding tools	17.09	**28.89**	**37.19**	14.82	2.01	0.00	M	8.29*	69.10	30.90	M
Sharing product links	**20.85**	**46.48**	19.85	7.29	2.76	2.76	I	25.63*	47.99	52.01	I
Shipment tracking	17.09	**18.34**	**41.21**	20.60	2.01	0.75	M	20.60*	78.89	21.11	M
Store locator	24.62	**32.66**	**26.88**	11.81	3.27	0.75	I	5.78[a]	63.32	36.68	A[b]
Wish list	21.36	**24.62**	**36.68**	12.56	4.02	0.75	M	12.06*	70.60	29.40	M
Supplementary features											
Augmented reality	**30.90**	**56.53**	1.01	4.77	2.51	4.27	I	25.63*	36.68	63.32	I
Chatbot	**22.11**	**57.04**	5.03	4.77	2.26	8.79	I	34.92*	31.91	68.09	I
Check and reserve	**37.69**	**39.95**	9.55	10.05	2.26	0.50	I	2.26[a]	57.29	42.71	A[b]
Click and collect	**32.91**	**45.98**	9.05	9.05	2.76	0.25	I	13.07*	51.01	48.99	I
Click and meet	**30.40**	**56.53**	3.02	6.28	2.76	1.01	I	26.13*	39.70	60.30	I
Comparison list	**52.26**	**18.09**	11.56	16.33	1.76	0.00	A	34.17*	80.15	19.85	A
Get the look	–	–	–	–	–	–	–	–	–	–	–
Inbox	**20.10**	**55.03**	5.53	5.03	2.76	11.56	I	34.92*	30.65	69.35	I
Live chat	**33.42**	**47.74**	6.03	5.53	2.51	4.77	I	14.32*	44.97	55.03	I
Loyalty program	**45.23**	**36.43**	6.03	8.79	2.51	1.01	A	8.79*	60.05	39.95	A
Online magazine	**14.32**	**69.85**	1.51	2.76	3.27	8.29	I	55.53*	18.59	81.41	I
Payment via app	**24.37**	**43.72**	16.83	9.05	2.01	4.02	I	19.35*	50.25	49.75	I
Personal product recommendations	**19.10**	**57.79**	3.27	2.51	2.76	14.57	I	38.69*	24.87	75.13	I
QR code scanner	**39.70**	**44.72**	4.27	6.03	3.77	1.51	I	5.03[a]	50.00	50.00	X[b]
Re-availability e-mail	**40.45**	**25.88**	15.33	14.32	2.76	1.26	A	14.57*	70.10	29.90	A
Size finder	–	–	–	–	–	–	–	–	–	–	–
Visual search	**41.96**	**45.48**	2.76	6.03	2.51	1.26	I	3.52[a]	50.75	49.25	A

Notes: bold = most and second most frequent classification; [a] (O + A + M) < > (I + R + Q) rule; [b] = change in category mode; * = categorization significant at a ten-percent level according to Lee and Newcomb (1997); X = mixed mode.

Data availability

Data will be made available on request.

References

Al-Nabhani, K., Wilson, A., McLean, G., 2021. Examining consumers' continuous usage of multichannel retailers' mobile applications. Psychol. Market. 39, 168–195. https://doi.org/10.1002/mar.21585.

Ampadu, S., Jiang, Y., Debrah, E., Antwi, C.O., Amankwa, E., Gyamfi, S.A., Amoako, R., 2022. Online personalized recommended product quality and e-impulse buying: a conditional mediation analysis. J. Retailing Consum. Serv. 64, 102789. https://doi.org/10.1016/j.jretconser.2021.102789.

Andrews, M., Goehring, J., Hui, S., Pancras, J., Thornswood, L., 2016. Mobile promotions: a framework and research priorities. J. Interact. Market. 34 (1), 15–24. https://doi.org/10.1016/j.intmar.2016.03.004.

Babin, B.J., Darden, W.R., Griffin, M., 1994. Work and/or fun: measuring hedonic and utilitarian shopping value. J. Consum. Res. 20 (4), 644–656. https://doi.org/10.1086/209376.

Baek, Y., 2013. Analysis of user's attitude toward apps, intention to use and continual consuming intention - focused on mobile commerce. International Journal of Contents 9 (4), 35–44. https://doi.org/10.5392/IJoC.2013.9.4.035.

Baier, D., Rese, A., 2020. How to increase multichannel shopping satisfaction? An adapted Kano based stage-gate approach to select new technologies. J. Retailing Consum. Serv. 56, 102172. https://doi.org/10.1016/j.jretconser.2020.102172.

Bauer, J.C., Linzmajer, M., Nagengast, L., Rudolph, T., D'Cruz, E., 2020. Gamifying the digital shopping experience: games without monetary participation incentives increase customer satisfaction and loyalty. Journal of Service Management 31 (3), 563–595. https://doi.org/10.1108/JOSM-10-2018-0347.

Beatty, S.E., Ferrell, M.E., 1998. Impulsive buying: modeling its precursors. J. Retailing 74 (2), 169–191. https://doi.org/10.1016/S0022-4359(99)80092-X.

Bellman, S., Potter, R.F., Treleaven-Hassard, S., Robinson, J.A., Varan, D., 2011. The effective-ness of brand mobile phone apps. J. Interact. Market. 25 (4), 191–200. https://doi.org/10.1016/j.intmar.2011.06.001.

Berger, C., Blauth, R., Boger, D., 1993. Kano's methods for understanding customer-defined quality. Center for Quality Management Journal 2 (4), 3–36.

Blom, A., Lange, F., Hess, R.L., 2021. Omnichannel promotions and their effect on customer satisfaction. Eur. J. Market. 55 (13), 177–201. https://doi.org/10.1108/EJM-12-2018-0866.

Boyd, D.E., Kannan, P.K., Slotegraaf, R.J., 2019. Branded apps and their impact on firm value: a design perspective. J. Market. Res. 56 (1), 76–88. https://doi.org/10.1177/0022243718820588.

Büttner, O.B., Florack, A., Göritz, A.S., 2014. Shopping orientation as a stable consumer disposition and its influence on consumers' evaluations of retailer communication. Eur. J. Market. 48 (5/6), 1026–1045. https://doi.org/10.1108/EJM-04-2012-0210.

Chang, C.-C., 2015. Exploring mobile application customer loyalty: the moderating effect of use contexts. Telecommun. Pol. 39 (8), 678–690. https://doi.org/10.1016/j.telpol.2015.07.008.

Chen, M.-C., Hsu, C.-L., Huang, C.-H., 2021. Applying the Kano model to investigate the quality of transportation services at mega events. J. Retailing Consum. Serv. 60, 102442. https://doi.org/10.1016/j.jretconser.2021.102442.

Chiu, T., Fang, D., Chen, J., Wang, Y., Jeris, C., 2001. A robust and scalable clustering algorithm for mixed type attributes in large database environment. In: Proceedings

of the Seventh ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, pp. 263–268.
De Keyser, A., Schepers, J., Konuş, U., 2015. Multichannel customer segmentation: does the after-sales channel matter? A replication and extension. Int. J. Res. Market. 32, 453–456. https://doi.org/10.1016/j.ijresmar.2015.09.005.
De Wulf, K., Odekerken-Schröder, G., Iacobucci, D., 2001. Investments in consumer relationships: a cross-country and cross-industry exploration. Journal of Marketing 65 (4), 33–50. https://doi.org/10.1509/jmkg.65.4.33.18386.
DelVecchio, D., Shanker Krishnan, H., Smith, D.C., 2007. Cents or percent? The effects of promotion framing on price expectations and choice. J. Market. 71, 158–170. https://doi.org/10.1509/jmkg.71.3.158.
Demko-Rihter, J., ter Halle, I., 2015. Revival of high street retailing – the added value of shopping apps. Amfiteatru Economic 17 (39), 632–645.
EHI Retail Institute, 2021. Top 100 onlineshops in deutschland. Retrieved September 22, 2022, from. https://www.ehi.org/de/top-100-umsatzstaerkste-onlineshops-in-deutschland/. (Accessed 6 December 2024).
Fong, D., 1996. Using the self-stated importance questionnaire to interpret Kano questionnaire results. Center for Quality of Management Journal 5 (3), 23.
Frasquet, M., Ievab, M., Ziliani, C., 2019. Understanding complaint channel usage in multichannel retailing. J. Retailing Consum. Serv. 47, 94–103. https://doi.org/10.1016/j.jretconser.2018.11.007.
Gilbride, T.J., Inman, J.J., Stilley, K.M., 2015. The role of within-trip dynamics in unplanned versus planned purchase behavior. J. Market. 79, 57–73. https://doi.org/10.1509/jm.13.0286.
Grewal, D., Gauri, D.K., Roggeveen, A.L., Sethuraman, R., 2021. Strategizing retailing in the new technology era. J. Retailing 97 (1), 6–12. https://doi.org/10.1016/j.jretai.2021.02.004.
Herhausen, D., Kleinlercher, K., Verhoef, P.C., Emrich, O., Rudolph, T., 2019. Loyalty formation for different customer journey segments. J. Retailing 95 (3), 9–29. https://doi.org/10.1016/j.jretai.2019.05.001.
Herzberg, F., 1966. Work and the Nature of Man. World Publishing Company, Cleveland, OH.
Herzberg, F., Mausner, B., Snyderman, B.B., 1959. The Motivation to Work. Wiley, New York, NY-London-Sydney.
Hoehle, H., Venkatesh, V., 2015. Mobile application usability: conceptualization and instrument development. MIS Q. 39 (2), 435–472. https://doi.org/10.25300/MISQ/2015/39.2.08.
Hsiao, C.-H., Chang, J.-J., Tang, K.-Y., 2016. Exploring the influential factors in continuance usage of mobile social apps: satisfaction, habit, and customer value perspectives. Telematics Inf. 33, 342–355. https://doi.org/10.1016/j.tele.2015.08.014.
Inman, J.J., Nikolova, H., 2017. Shopper-facing retail technology: a retailer adoption decision framework incorporating shopper attitudes and privacy concerns. J. Retailing 93 (1), 7–28. https://doi.org/10.1016/j.jretai.2016.12.006.
Iyer, P., Davari, A., Mukherjee, A., 2018. Investigating the effectiveness of retailers' mobile applications in determining customer satisfaction and repatronage intentions? A congruency perspective. J. Retailing Consum. Serv. 44, 235–243. https://doi.org/10.1016/j.jretconser.2018.07.017.
Kano, N., Seraku, N., Takahashi, F., Tsuji, S., 1984. Attractive quality and must-be quality. Journal of the Japanese Society for Quaility Control 14 (2), 39–48.
Kim, S.C., Yoon, D., Han, E.K., 2016. Antecedents of mobile app usage among smartphone users. J. Market. Commun. 22 (6), 653–670. https://doi.org/10.1080/13527266.2014.951065.
Kim, M., Kim, J., Choi, J., Trivedi, M., 2017. Mobile shopping through applications: understanding application possession and mobile purchase. J. Interact. Market. 39, 55–68. https://doi.org/10.1016/j.intmar.2017.02.001.
Kim, S.-A., Park, S., Kwak, M., Kang, C., 2025. Examining product quality and competitiveness via online reviews: an integrated approach of importance performance competitor analysis and Kano model. J. Retailing Consum. Serv. 2, 104135. https://doi.org/10.1016/j.jretconser.2024.104135.
Konuş, U., Verhoef, P.C., Neslin, S.A., 2008. Multichannel shopper segments and their covariates. J. Retailing 84 (4), 398–413. https://doi.org/10.1016/j.jretai.2008.09.002.
Kranzbühler, A.M., Kleijnen, M.H., Verlegh, P.W., 2019. Outsourcing the pain, keeping the pleasure: effects of outsourced touchpoints in the customer journey. J. Acad. Market. Sci. 47 (2), 308–327. https://doi.org/10.1007/s11747-018-0594-5.
Lee, M.C., Newcomb, J., 1997. Applying the Kano methodology to meet customer requirements: NASA's microgravity science program. Qual. Manag. J. 4 (3), 95–106. https://doi.org/10.1080/10686967.1997.11918805.
Li, X., Zhao, X., Xu, W., Pu, W., 2020. Measuring ease of use of mobile applications in e-commerce retailing from the perspective of consumer online shopping behaviour patterns. J. Retailing Consum. Serv. 55, 102093. https://doi.org/10.1016/j.jretconser.2020.102093.
Liang, T.-P., Lai, H.-J., Ku, Y.-C., 2007. Personalized content recommendation and user satisfaction: theoretical synthesis and empirical findings. J. Manag. Inf. Syst. 23 (3), 45–70. https://doi.org/10.2753/MIS0742-1222230303.
Liu, P., Li, M., Dai, D., Guo, L., 2021. The effects of social commerce environmental characteristics on customers' purchase intentions: the chain mediating effect of customer-to-customer interaction and customer-perceived value. Electron. Commer. Res. Appl. 48, 101073. https://doi.org/10.1016/j.elerap.2021.101073.
Lo, L.Y.-S., Lin, S.-W., Hsu, L.-Y., 2016. Motivation for online impulse buying: a two-factor theory perspective. Int. J. Inf. Manag. 36, 759–772. https://doi.org/10.1016/j.ijinfomgt.2016.04.012.
Maduku, D.K., Thusi, P., 2023. Understanding consumers' mobile shopping continuance intention: new perspectives from South Africa. J. Retailing Consum. Serv. 70, 103185. https://doi.org/10.1016/j.jretconser.2022.103185.
Nakano, S., Kondo, F.N., 2018. Customer segmentation with purchase channels and media touchpoints using single source panel data. J. Retailing Consum. Serv. 41, 142–152. https://doi.org/10.1016/j.jretconser.2017.11.012.
Natarajan, T., Balasubramanian, S.A., Kasilingam, D.L., 2017. Understanding the intention to use mobile shopping applications and its influence on price sensitivity. J. Retailing Consum. Serv. 37, 8–22. https://doi.org/10.1016/j.jretconser.2017.02.010.
Ortlinghaus, A., Zielke, S., 2019. Designing multi-channel technologies – the mediating role of risk perceptions. Int. Rev. Retail Distrib. Consum. Res. 29 (5), 518–536. https://doi.org/10.1080/09593969.2019.1664610.
Ortlinghaus, A., Zielke, S., Dobbelstein, T., 2019. The impact of risk perceptions on the attitude toward multi-channel technologies. Int. Rev. Retail Distrib. Consum. Res. 29 (3), 262–284. https://doi.org/10.1080/09593969.2019.1599983.
Park, S., Lee, D., 2017. An empirical study on consumer online shopping channel choice behavior in omni-channel environment. Telematics Inf. 34 (8), 1398–1407. https://doi.org/10.1016/j.tele.2017.06.003.
Peng, K.-F., Chen, Y., Wen, K.-W., 2014. Brand relationship, consumption values and branded app adoption. Ind. Manag. Data Syst. 114 (8), 1131–1143. https://doi.org/10.1108/IMDS-05-2014-0132.
Qin, H., Osatuyi, B., Xu, L., 2021. How mobile augmented reality applications affect continuous use and purchase intentions: a cognition-affect-conation perspective. J. Retailing Consum. Serv. 63, 102680. https://doi.org/10.1016/j.jretconser.2021.102680.
Rese, A., Schlee, T., Baier, D., 2019. The need for services and technologies in physical fast fashion stores: generation Y's opinion. Journal of Marketing Management 35 (15–16), 1437–1459. https://doi.org/10.1080/0267257X.2019.1665087.
Roggeveen, A.L., Sethuraman, R., 2020. Customer-interfacing retail technologies in 2020 & beyond: an integrative framework and research directions. J. Retailing 96 (3), 299–309. https://doi.org/10.1016/j.jretai.2020.08.001.
Rook, D.W., Fisher, R.J., 1995. Normative influences on impulsive buying behavior. J. Consum. Res. 22 (3), 305–313. https://doi.org/10.1086/209452.
Sarkar, S., Khare, A., 2018. Influence of expectation confirmation, network externalities, and flow on use of mobile shopping apps. Int. J. Hum. Comput. Interact. 35 (16), 1449–1460. https://doi.org/10.1080/10447318.2018.1540383.
Sauerwein, E., 1999. Experiences with the reliability and validity of the Kano-method: comparison to alternate forms of classification of product requirements. In: In Transactions of the 11th Symposium on Quality Function Deployment. QFD Institute, pp. 416–429.
Schneider, P.J., Zielke, S., 2020. Searching offline and buying online – an analysis of showrooming forms and segments. J. Retailing Consum. Serv. 52, 101919. https://doi.org/10.1016/j.jretconser.2019.101919.
Shankar, V., Venkatesh, A., Hofacker, C., Naik, P., 2010. Mobile marketing in the retailing environment: current insights and future research avenues. J. Interact. Market. 24 (2), 111–120. https://doi.org/10.1016/j.intmar.2010.02.006.
Shokouhyar, S., Shokoohyar, S., Safari, S., 2020. Research on the influence of after-sales service quality factors on customer satisfaction. J. Retailing Consum. Serv. 56, 102139. https://doi.org/10.1016/j.jretconser.2020.102139.
Sinemus, K., Zielke, S., Dobbelstein, T., 2022. Shopping app features: their impact on customer satisfaction and loyalty. Int. Rev. Retail Distrib. Consum. Res. 32 (4), 423–449. https://doi.org/10.1080/09593969.2022.2109189.
Statista, 2024. Nutzung von Apps von Einzelhandelsunternehmen nach Altersgruppen in Deutschland 2022. https://de.statista.com/statistik/daten/studie/1325067/umfrage/nutzung-von-apps-von-einzelhandelsunternehmen-nach-altersgruppen-in-deutschland/. (Accessed 6 December 2024).
Statistisches Bundesamt, 2024. Bevölkerung: deutschland, stichtag, altersjahre. https://www-genesis.destatis.de/datenbank/online/table/12411-0005/search/s/MTI0MTEtMDAwNQ. (Accessed 6 December 2024).
Sundström, M., Hjelm-Lidholm, S., Radon, A., 2019. Clicking the boredom away – exploring impulse fashion buying behavior online. J. Retailing Consum. Serv. 47, 150–156. https://doi.org/10.1016/j.jretconser.2018.11.006.
Taylor, D.G., Levin, M., 2014. Predicting mobile app usage for purchasing and information-sharing. Int. J. Retail Distrib. Manag. 42 (8), 759–774. https://doi.org/10.1108/IJRDM-11-2012-0108.
Tong, S., Luo, X., Xu, B., 2020. Personalized mobile marketing strategies. J. Acad. Market. Sci. 48, 64–78. https://doi.org/10.1007/s11747-019-00693-3.
Venkatesh, V., Thong, J.Y.L., Xu, X., 2012. Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. MIS Quarterly 36 (1), 157–178. https://doi.org/10.2307/41410412.
Verhoef, P.C., Kannan, P.K., Inman, J.J., 2015. From multi-channel retailing to omni-channel retailing introduction to the special issue on multi-channel retailing. J. Retailing 91 (2), 174–181. https://doi.org/10.1016/j.jretai.2015.02.005.
Wang, Y.J., Minor, M.S., Wei, J., 2011. Aesthetics and the online shopping environment: understanding consumer responses. J. Retailing 87, 46–58. https://doi.org/10.1016/j.jretai.2010.09.002.
Wenninger, A., Rau, D., Röglinger, M., 2022. Improving customer satisfaction in proactive service design. Electron. Mark. 32, 1399–1418. https://doi.org/10.1007/s12525-022-00565-9.
Zare Ebrahimabad, F., Yazdani, H., Hakim, A., Asarian, M., 2024. Augmented reality versus web-based shopping: how does AR improve user experience and online purchase intention. Telematics and Informatics Reports 15, 100152. https://doi.org/10.1016/j.teler.2024.100152.

Zhang, L., Xuan, Y., Li, Z., Gao, P., Zheng, Y., 2024. How to obtain customer requirements for each stage of the product life cycle from online reviews: using mobile phones as an example. J. Retailing Consum. Serv. 80, 103928. https://doi.org/10.1016/j.jretconser.2024.103928.

Zhao, Z., Balagué, C., 2015. Designing branded mobile apps: fundamentals and recommendations. Bus. Horiz. 58, 305–315. https://doi.org/10.1016/j.bushor.2015.01.004.

Zhao, M., Dholakia, R.R., 2009. A multi-attribute model of web site interactivity and customer satisfaction - an application of the Kano model. Manag. Serv. Qual. 19 (3), 286–307. https://doi.org/10.1108/09604520910955311.

Zielke, S., 2008. Exploring asymmetric effects in the formation of retail price satisfaction. J. Retailing Consum. Serv. 15, 335–347. https://doi.org/10.1016/j.jretconser.2007.08.005.